

Credit Card Status Report

Developed an end-to-end **Credit Card Customer & Transaction Analytics Dashboard** in **Power BI**, integrating customer and transaction datasets to provide actionable business insights.

Tools Used: **Power BI, DAX, Power Query, Data Modeling, Data Visualization, Business Intelligence, Customer Analytics.**

Project Approach:

- Built and optimized the **ETL pipeline** using **Power Query**, including data cleansing, transformation, data type standardization, and data modeling.
- Designed a star-schema data model by joining customer and transaction tables using **Client_Num**, improving reporting efficiency and data accuracy.
- Created advanced **DAX measures** for Revenue, Profit, Transaction Frequency, At-Risk Revenue, Week-over-Week (WoW) Growth, Customer Satisfaction, and High-Income Customer Analysis.
- Developed calculated columns for **customer segmentation** based on Age Group and Income Group to support targeted business analysis.
- Designed interactive dashboards **Credit Card Transaction Report** and **Credit Card Customer Report** with dynamic slicers, drill-through functionality, and page navigation to enhance user-driven analysis.
- Analysed over **10,000+ customer records** to identify revenue-driving customer segments, spending behavior patterns, and delinquency risks.
- Identified **₹3M+ At-Risk Revenue** and high-risk customer segments through revolving balance and utilization ratio analysis.
- Performed customer profitability and acquisition analysis, highlighting opportunities to improve card activation rates and marketing effectiveness.
- Generated business recommendations that supported customer retention, revenue growth, risk monitoring, and targeted marketing strategies.
- Resolved data quality issues involving regional date formats using Power Query locale settings, ensuring accurate weekly trend reporting.

Key Insights

- Bill payment transactions emerged as the largest contributor to overall credit card revenue.
- Blue Card customers generated **83.49% of total revenue**, making them the most valuable customer segment.
- Swipe transactions accounted for **66.93% of total transaction volume**, indicating strong preference for traditional payment methods.
- Revenue exhibited seasonal peaks during **January, April, July, October, and December**, with July generating the highest revenue of **₹5.65M**.
- Week 53 recorded the highest **WoW Revenue Growth of 28.8%**, indicating a potential promotional or seasonal impact.
- Identified high-risk customer segments with elevated **credit utilization ratios and revolving balances**, particularly among delinquent Gold Card customers.

- At-Risk Revenue totalled approximately **₹3M**, highlighting potential exposure to future defaults.
- Customers aged **46+** contributed **55.28%** of total revenue, making them the most profitable age segment.
- Low-income customers generated a significant share of revenue despite lower earning capacity, indicating strong engagement with credit products.
- Analysis revealed **568 delinquent Blue Card customers** with substantial revolving balances requiring closer monitoring.
- Female customers represented the majority of the customer base, accounting for **58% of total customers**.
- Nearly **74.3% of customer acquisition spending** was associated with customers who did not activate their cards within 30 days.

Recommendations

- Strengthen retention and loyalty programs for high-value Blue Card customers to sustain revenue growth.
- Conduct root-cause analysis on revenue spikes and dips to identify successful campaigns and improve future planning.
- Implement proactive monitoring and early-warning systems for customers with high utilization ratios and delinquent accounts.
- Prioritize recovery strategies for the **₹3M At-Risk Revenue** portfolio to reduce potential credit losses.
- Develop targeted marketing campaigns aimed at increasing engagement among customers aged **26-35**, a relatively underperforming segment.
- Introduce tailored offers and premium benefits to attract higher-spending customers from the high-income group.
- Improve customer onboarding and follow-up processes to increase card activation rates within the first 30 days.
- Launch personalized payment reminders and financial wellness programs for customers exhibiting delinquency risk.
- Promote digital payment methods to diversify transaction channels and reduce dependency on swipe transactions.
- Leverage customer segmentation insights to design more effective cross-selling and upselling strategies across card categories and customer demographics.

Bottom of Form

Challenge:

Date parse error appeared in CreditCard table as Week_start_date column had date according to regional setting of United States i.e. MM-DD-YYYY but power bi had regional settings for India.

Solution: In power query right clicked week_start_date column and

Clicked change type – using locale -

Data type: Date

Locale: English (United States).